



IBM Developer  
SKILLS NETWORK

# Winning Space Race with Data Science

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# Outline

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- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

# Executive Summary

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## Context

The commercial space industry is expanding, with companies aiming to reduce orbital access costs. SpaceX leads by reusing Falcon 9 boosters. Competitor SpaceY is evaluating booster recovery by analyzing SpaceX data to predict recovery success.

## Applied Methodologies

- **Data Collection** – collected SpaceX launch data using SpaceX's public REST API & web scraping its Wikipedia page.
- **Data Wrangling** – Cleaned and processed the collected data.
- **Exploratory Data Analysis(EDA)** - performed EDA to identify features influencing booster recovery and selected those features for predictive analysis.
- **Build Interactive Maps and Visual Analytics Dashboard**
  - built interactive maps to visually compare launch site locations, view success/failure rates & proximity to key infrastructure for each launch site.
  - built interactive dashboard analytics to help identify optimal launch sites and payloads with the highest likelihood of success.
- **Predictive Analysis** – built 4 types of classification models on subset of launch data(selected features) and picked the best model with the highest accuracy score.

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## Summary of Findings

- Key factors that influence booster recovery include:
  - Flight Number
  - Payload mass
  - Orbit type
  - Launch Site
  - Landing method
  - Booster version
- Model outcomes - **Decision Tree Classifier model performed best & demonstrated:**
  - High accuracy of ~**88.9%** in predicting booster landing outcomes
  - Consistent performance across validation sets
  - Enhanced interpretability through analysis of feature importance

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- **Key Takeaways:**

- Heavier payloads and missions targeting higher-energy orbits greatly decrease the chances of booster recovery.
- Recovery success rates improved significantly after year 2015.
- Incorporating recovery probability into launch cost estimates allows SpaceY to enhance accuracy of its pricing models.

- **Business implications** - The predictive model and dashboards enable SpaceY to:

- Pinpoint mission profiles where reusable boosters deliver the most cost savings
- Provide more precise mission cost forecasts
- Compare cost efficiencies against SpaceX benchmarks
- Facilitate competitive proposals and strategic decision-making

# Introduction

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## Project Background and Context

- Rapid advancements in reusable rocket technology are driving growth in commercial space sector.
- SpaceX sets new cost benchmark by reusing Falcon 9 boosters, reducing launch costs from industry average of \$165 million to about \$62 million.
- To compete effectively, SpaceY must understand key factors influencing booster recovery for accurate cost forecast and maintain competitive pricing.
- This project uses publicly available SpaceX launch data to develop a predictive model that supports strategic decision-making for SpaceY's launch operations.

## Key Questions to Explore

- Which mission factors influence first-stage landing attempts and success?
- Can booster recovery be predicted from mission profiles?
- Can SpaceX's launch patterns inform SpaceY's pricing, planning, and competitive strategy?



Section 1

# Methodology

# Methodology

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## Methodologies Overview

- **Data Collection** – SpaceX Launch Data was collected via public REST API calls & web scraping Falcon 9 wiki page and responses converted to Pandas data frame which is then filtered & cleaned.
- **Data Wrangling** – New landing outcome label 'Class' is added as column to Pandas data frame and filled with values using 'outcome' column.
- **Perform Exploratory Data Analysis (EDA)** – Plotted graphs(using Seaborn and Matplotlib python libraries) and ran SQL queries to get insights into features affecting success rate & selected those features for further predictive analysis.
- **Perform interactive visual analytics** using Folium and Plotly Dash - Built interactive maps to visually compare launch site locations, their proximities to key infrastructure. Built a Analytics Dashboard that helps identify optimal launch site and payloads with the highest likelihood of success
- **Perform predictive analysis** using classification models – built models to predict booster recovery, evaluate performance & find best model.



# Data Collection

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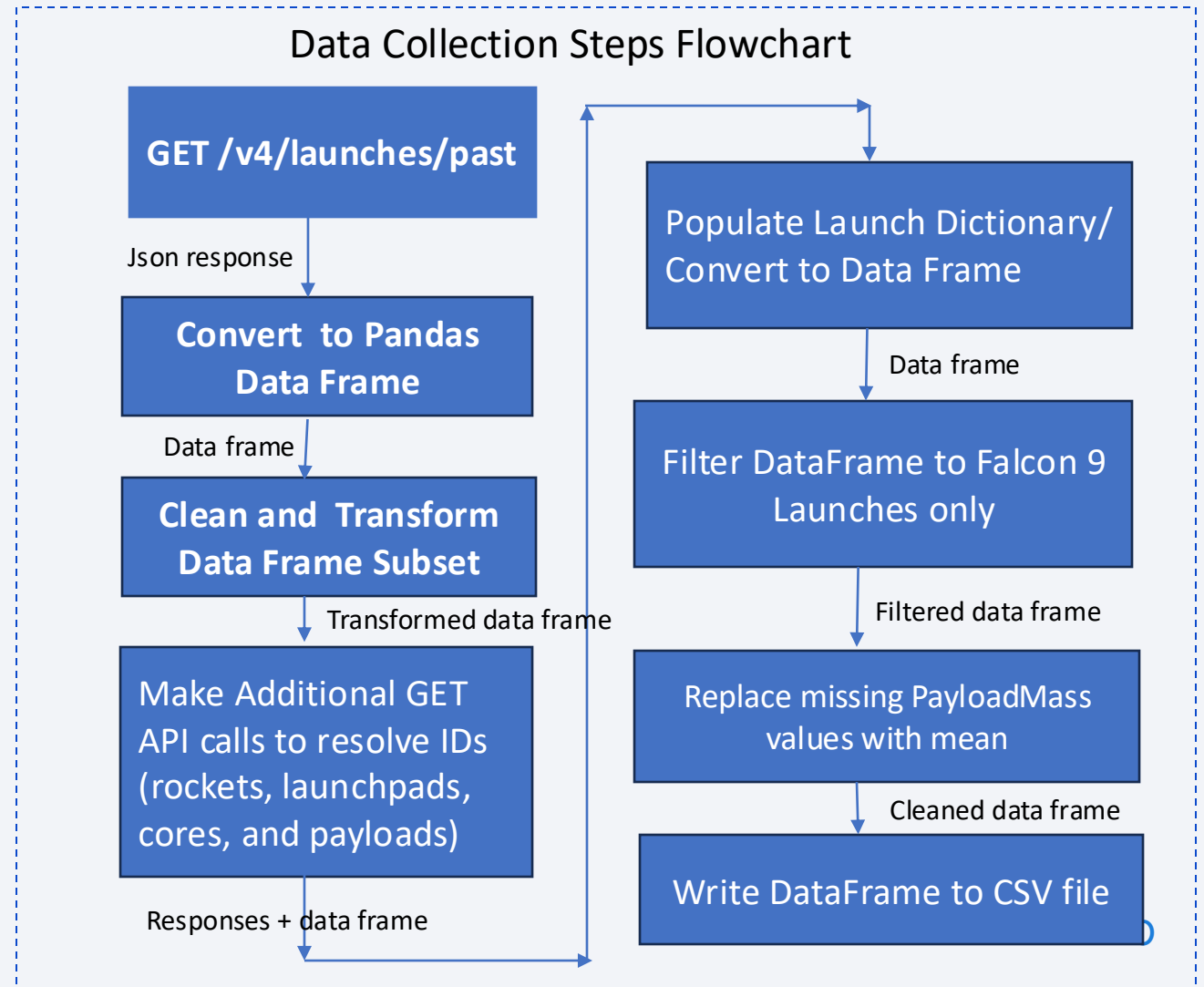
- SpaceX launch data is collected by
  - Fetching data from SpaceX public REST API calls using Python requests library.
  - Web Scraping SpaceX Falcon 9 Wiki Pages using BeautifulSoup Python library.
- API response is normalized (using json\_normalize) and ID fields (rocket, payload, cores etc.) are replaced with detailed information.
- Data is filtered to contain Falcon 9 launches only
- Lastly, data is cleaned by imputing missing PayloadMass with mean and retaining NULLs in LandingPad

# Data Collection – SpaceX API

Base URL: [api.spacexdata.com/v4/](https://api.spacexdata.com/v4/)

- 1) Make GET request to API `/launches/past` (json response converted to pandas data frame)
- 2) Clean data frame - Remove rows w/ multiple cores, payloads & replace lists w/ single value
- 3) Resolve IDs - Make more GET requests
  - `/rockets/`      ▪ `/launchpads/`
  - `/payloads/`    ▪ `/cores/`
- 4) Use resolved data with original launch data to create pandas data frame (with launch dictionary as intermediate data structure)
- 5) Filter data frame to 'Falcon 9' launches and handle missing values.

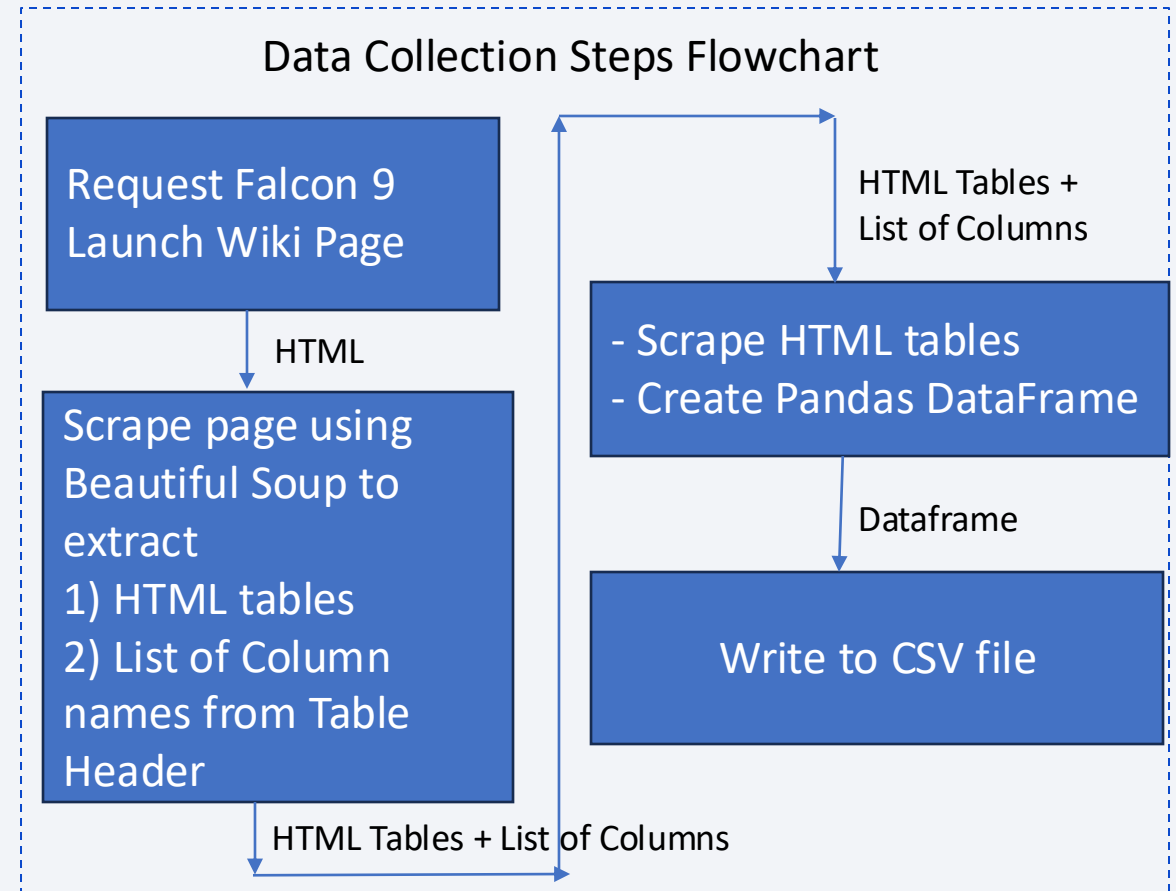
Link to Code on Github is [here](#)



# Data Collection - Scraping

- 1) Make GET Request for the SpaceX Falcon 9 Launches [Wikipedia Page](#)
- 2) Scrape the wiki page using **Beautiful Soup** Python Library to extract HTML tables and list of column names from those HTML table headers
- 3) Scrape the HTML tables that were extracted and use that data long with the column names to create a pandas data frame

Link to Code on GitHub is [here](#)

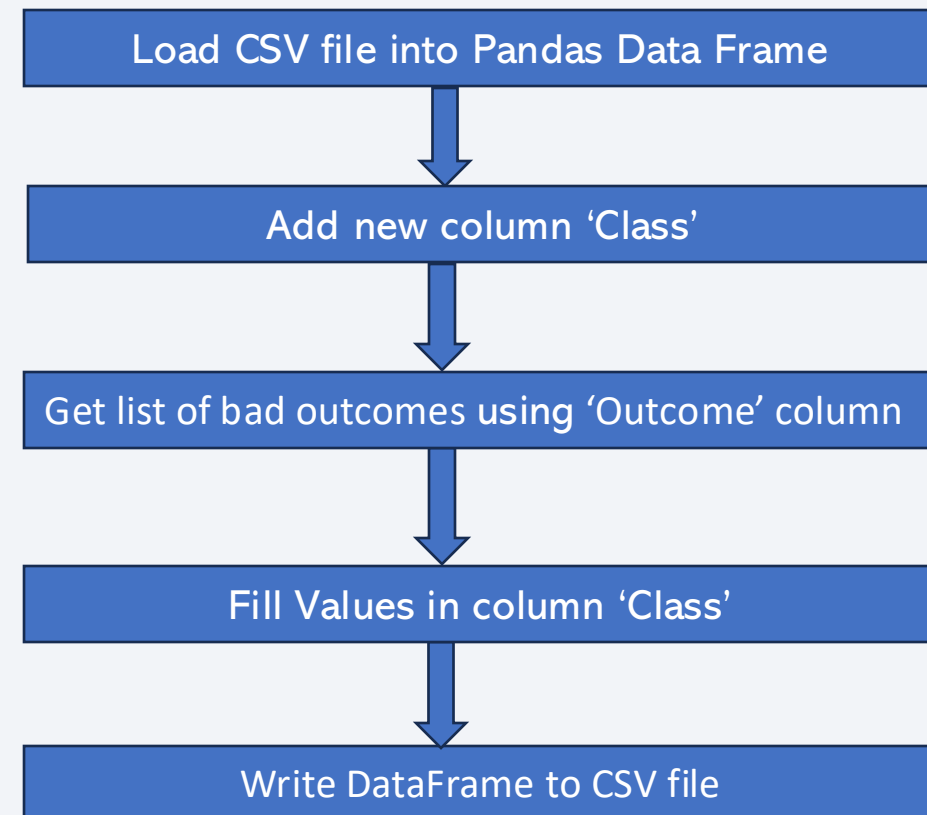


# Data Wrangling

- Performed EDA listed below on data frame(data collection phase) to find patterns and determine label for training supervised models.
  - # of launches on each site
  - # and occurrence of each orbit
  - # and occurrence of mission outcome
- Created landing outcome label 'Class' from 'outcome' column – fill value 0 if 'Outcome' in bad outcomes & 1 otherwise.
  - True ASDS
  - True RTLS
  - True Ocean

Link to Code on GitHub is [here](#)

## PROCESS FLOWCHART



# EDA with Data Visualization

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- Graphs (listed below) were plotted to gain preliminary insights into factors affecting success rate

- Categorical plot of **Flight Number vs Payload Mass**
- Categorical plot of **Payload Mass vs Launch Site**
- Categorical plot of **Flight Number vs Orbit Type**
- Line Chart of **Year vs Average Success Rate**
- Categorical plot of **Flight Number vs Launch Site**
- Bar Chart of **Success Rate vs Orbit Type**
- Categorical Plot of **PayloadMass vs Orbit Type**

- Based on insights from graphs(above) we selected features (below) used in success prediction

- FlightNumber
- PayloadMass
- Orbit
- LaunchSite
- Flights
- GridFins
- Reused
- Legs
- LandingPad
- Block
- ReusedCount
- Serial

- Converted categorical columns to numeric ( used pandas **get\_dummies**) & Cast all numeric columns to float64

Link to Code on GitHub is [here](#)



# EDA with SQL

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- Performed EDA Using SQL to gain better understanding of SpaceX data.
- Loaded dataset into DB2 database.
- Executed SQL Queries below to answer get more insights into SpaceX data

|                                                                                                                         |                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| • Names of unique launch sites in the space mission?                                                                    | • 5 records where launch sites begin with string 'CCA' ?                                          |
| • Total payload mass carried by boosters launched by NASA (CRS) ?                                                       | • Average payload mass carried by booster version F9 v1.1 ?                                       |
| • When first successful landing outcome in ground pad was achieved. ?                                                   | • Booster names with success in drone ship having payload mass > 4000 and <6000. ?                |
| • Total successful and failure mission outcomes. ?                                                                      | • Booster versions that carried maximum payload mass. ?                                           |
| • Display records in year 2015 with month name, failed landing_outcomes in drone ship, booster_versions, launch_site. ? | • Rank count of landing outcomes between the date 2010-06-04 and 2017-03-20in descending order. ? |

[Link to Code on GitHub is here](#)

# Build Interactive Map with Folium

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## Created Interactive Map using -

- Folium Map object – focused on geographic area containing SpaceX launch sites.
- Marker objects – placed on map to pinpoint exact locations of launch sites.
- Proximity markers (including nearby city, railway, highway, and coastline).
- Enabled user interactions like zooming, panning, and viewing marker details.

## Purpose?

- Visually compare launch site locations and surrounding geographic features.
- Assist users to recognize patterns e.g. site proximity to key infrastructure.
- Support analysis to select optimal launch site based on spatial relationships.
- Offer interactive experience to convey insights better than static maps/charts.
- Gain insights to enhance a Plotly Dash dashboard application

Link to Code on GitHub is [here](#)

# Build Dashboard with Plotly Dash

## Plots/Graphs

| Pie Chart – Launch Success Counts                                                                                                                                                                                                                                                 | Scatter Plot – Payload vs. Launch Outcome                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>Displays total # of successful launches per launch site.</li></ul> <p>Success vs. Failure pie chart for any selected launch site</p> <ul style="list-style-type: none"><li>Helps quickly compare performance across launch sites.</li></ul> | <ul style="list-style-type: none"><li>Plots payload mass (kg) against launch success or failure.</li><li>Points colored by booster version so users can compare performance.</li><li>Helps explore relation b/w payload size, booster type, and launch outcome.</li></ul> |

## Interactions Added

| Launch Site Dropdown Menu                                                                                                                                                                                                  | Payload Range Slider                                                                                                                                                                                                                                  | Interactive Hover-Zoom-Pan                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>Allows selection of 'All'/any specific site</li><li>Updates both pie chart &amp; scatter plot dynamically.</li><li>Enables flexible comparisons and site-specific analysis</li></ul> | <ul style="list-style-type: none"><li>Allows filtering launches by payload mass.</li><li>Updates scatter plot to show only launches in selected payload range.</li><li>Helps users identify how payload sizes influence success likelihood.</li></ul> | <ul style="list-style-type: none"><li>Users can hover to see detailed data values.</li><li>Zooming &amp; panning allow deeper exploring of dense areas on scatter plot.</li></ul> |

# Build a Dashboard with Plotly Dash

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## Purpose of Plots and Interactions ?

- Help uncover patterns hidden in static charts.
- Pie chart highlights top launch sites quickly.
- Scatter plot shows links b/w payload mass, booster type, and launch success.
- Site dropdown enables easy site comparisons without clutter.
- Range slider focuses on trends in specific payload ranges.
- Interactive elements allow real-time data analysis for better clarity and engagement.

Link to Code on GitHub is [here](#)

# Predictive Analysis (Classification)

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- Developed ML pipeline that predicts if Falcon 9 first stage landing is success
- Process involved 4 Key Phases:

## 1. Building the Models

- Data set preprocessed - **standardized** so all features are on same scale.
- Data set split into training and testing sets using **train\_test\_split**
- Four classification algorithms were trained
  - Logistic Regression
  - Support Vector Machine
  - Decision Tree Classifier
  - K-Nearest Neighbors (KNN)

## 2. Evaluating Initial Model Performance

- Each model trained on training data using default setting
- Use accuracy scores to evaluate perf. and establish baselines across 4 algorithms

# Predictive Analysis (Classification)

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## 3. Improving Models with Hyperparameter Tuning

- Grid Search employed for each algorithm to determine optimal hyperparameters.
- Enables systematic testing of parameter combinations to fine-tune models for maximum performance.
- Model accuracy reassessed on training set using best hyperparameter settings.

## 4. Selecting Best Performing Model

- Model with highest training accuracy chosen as top performer.
- Confusion Matrix created using test dataset to evaluate the final model's performance and identify prediction errors.

Link to Code on GitHub is [here](#)



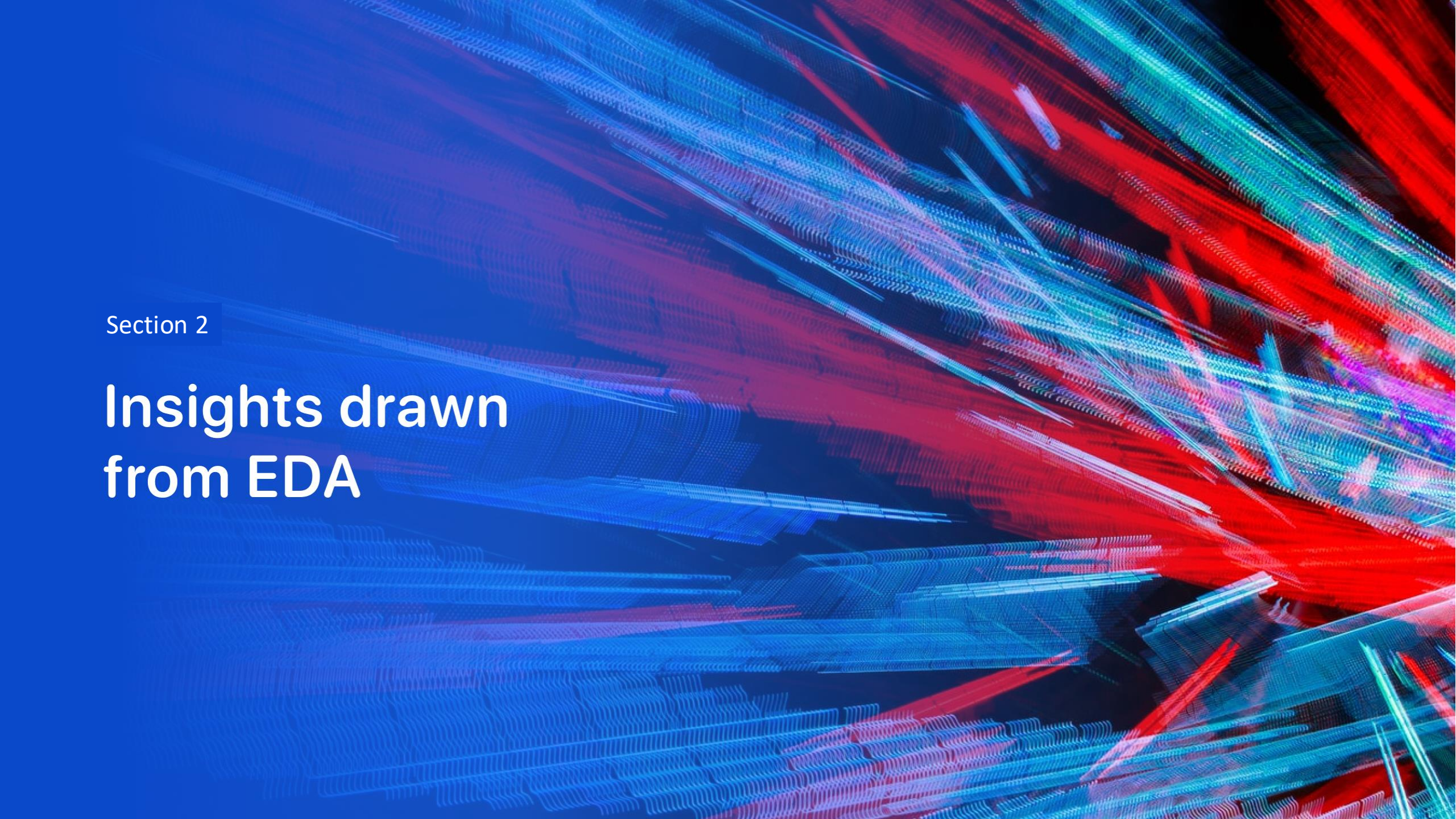
# Results

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## Overview

- Exploratory Data Analysis Findings
  - Key factors influencing booster recovery include – Flight #, Payload Mass, Orbit Type, Launch Site, Landing method and Booster Version
  - Missions targeting Low Energy Orbits (LEO, VLEO, SSO, PO) have higher chance of booster recovery.
- Interactive Maps & Analytics Dashboard Results
  - KSC LC-39A Launch site has the highest launch success rate of about 76%
  - Payload range 3K-4K (Kgs) has best success rate of 70% but above 6K success rate drops to 16%.
  - F9 Booster Version FT has best launch success rate followed by B4 and B5 – v1.0 & v1.1 have lowest
- Decision Tree Classifier Model performed better than its counterparts on both training & testing sets with accuracy scores of 0.875 & 0.888 respectively.



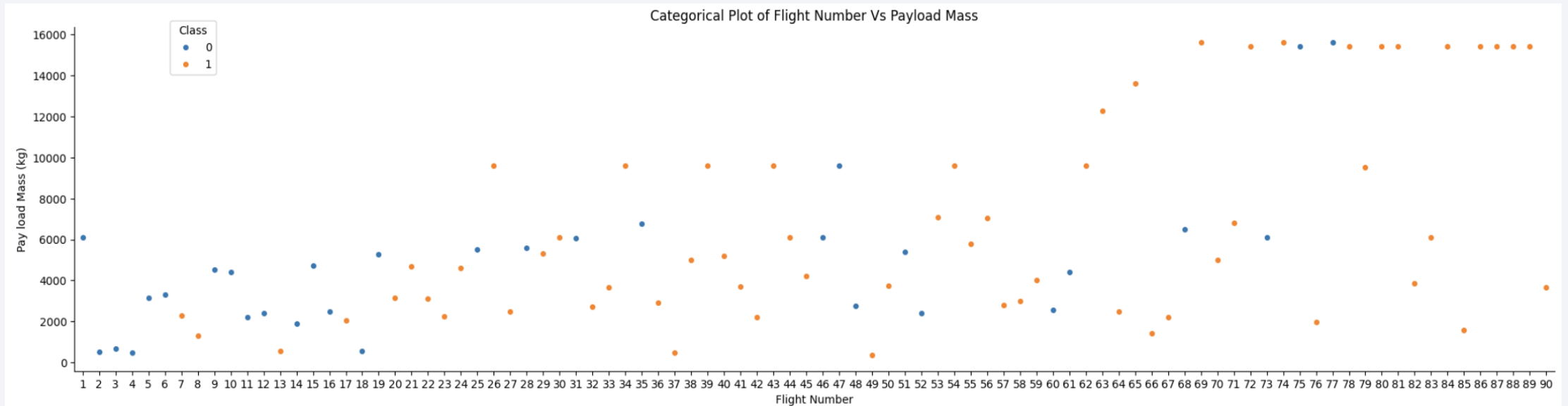
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

Section 2

# Insights drawn from EDA

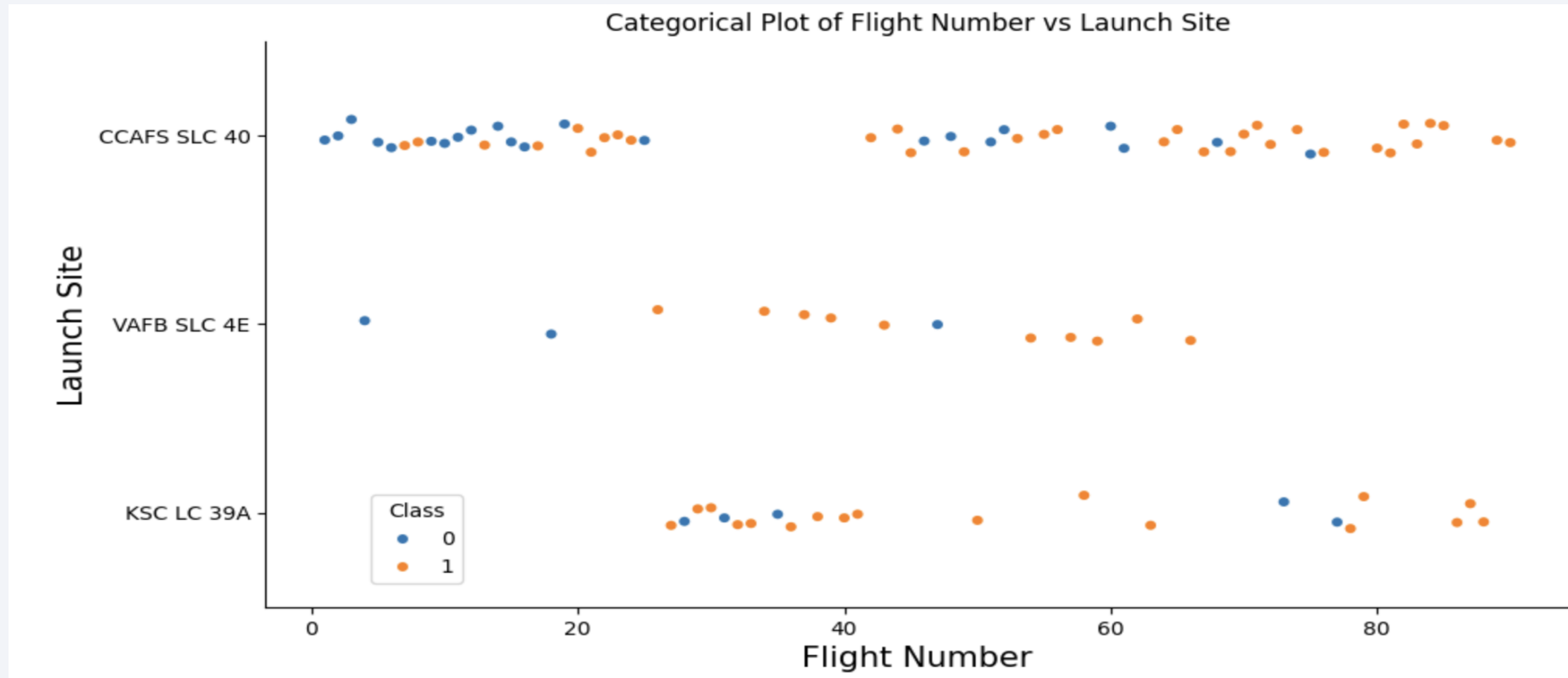


# Flight Number vs. Payload Mass



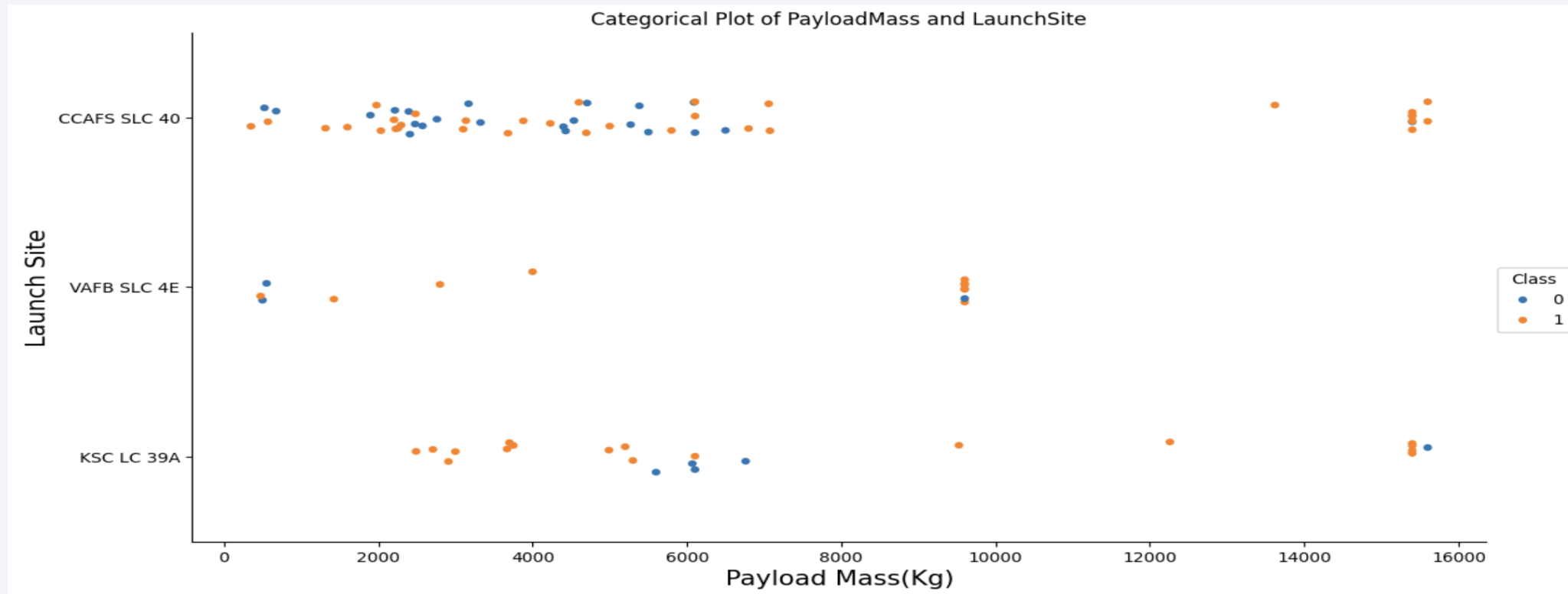
- As flight # increases(continuous launch attempts) - first stage is more likely to land successfully.
- The payload mass also appears to be a factor - even with more massive payloads( >6000kgs) first stage often returns successfully

# Flight Number vs. Launch Site



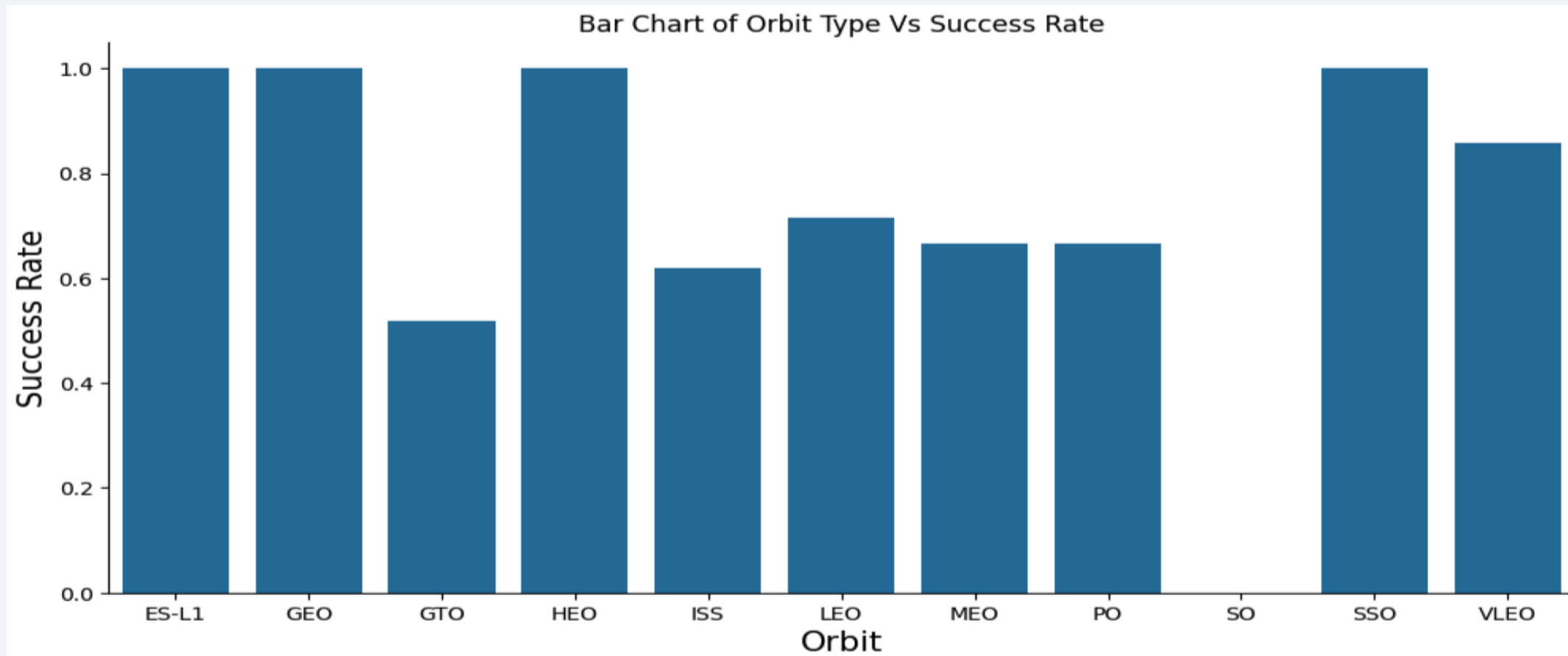
- For all launch sites as flight # increases first stage is more likely to land successfully.
- 'VAFB SLC 4E' and 'KSC LC 39A' sites have higher probability of successful landing
- 'CCAFS SLC 40' success probability is low for flight #s < 20 but goes up for > 20

# Payload vs. Launch Site



- 'VAFB SLC 4E' - No rockets were launched with payloads  $> 10,000$  kg.
- 'CCAFS SLC 40' - seems good at launching rockets w/ payloads  $> 10,000$ kg - however sample size is too small.
- 'KSC LC 39A' - very good success rate with wide range of payload mass b/w 2,000 –16000 Kg.

# Success Rate vs. Orbit Type

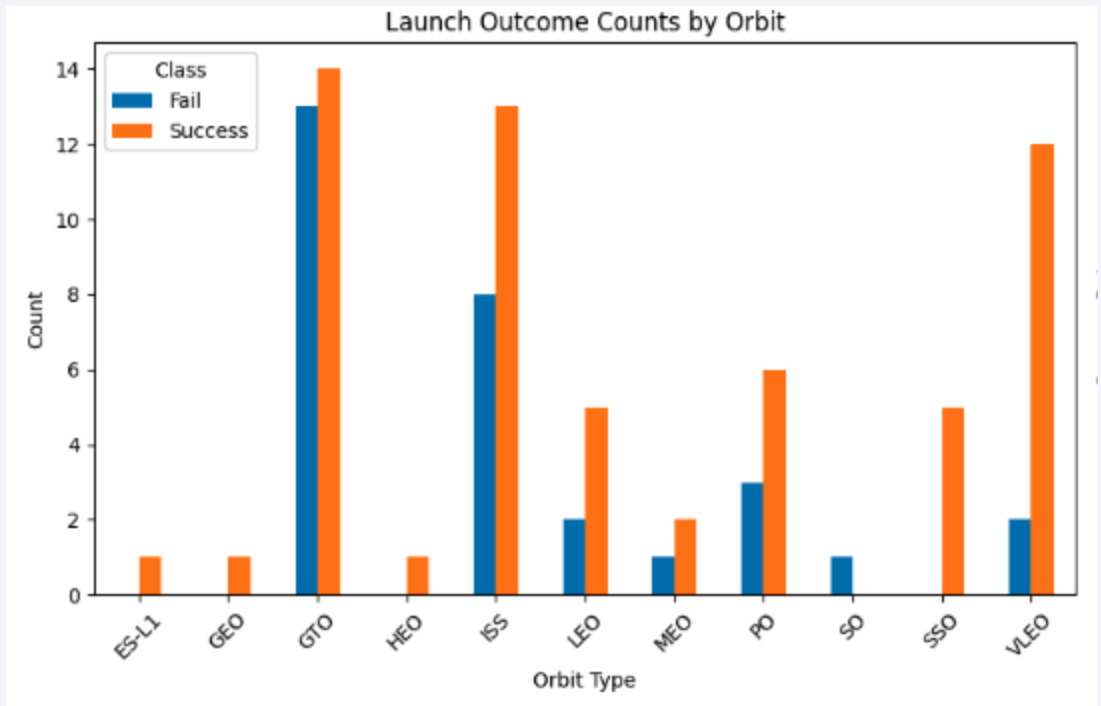


- Orbits with Highest Success Rates appear to be : **ES-L1, GEO, HEO & SSO**

\* However note that sample sizes for ES-L1, GEO & HEO is just 1 and if the one sample succeeds it doesn't mean that these orbit have 100% success rates. So we have to look at orbits with larger sample sizes

# Launch Outcome Counts by Orbit Type

| Orbit | Fail | Success | Total | Success_Rate |
|-------|------|---------|-------|--------------|
| ES-L1 | 0    | 1       | 1     | 1.0          |
| GEO   | 0    | 1       | 1     | 1.0          |
| GTO   | 13   | 14      | 27    | 0.518        |
| HEO   | 0    | 1       | 1     | 1.0          |
| ISS   | 8    | 13      | 21    | 0.619        |
| LEO   | 2    | 5       | 7     | 0.714        |
| MEO   | 1    | 2       | 3     | 0.667        |
| PO    | 3    | 6       | 9     | 0.667        |
| SO    | 1    | 0       | 1     | 0.0          |
| SSO   | 0    | 5       | 5     | 1.0          |
| VLEO  | 2    | 12      | 14    | 0.857        |



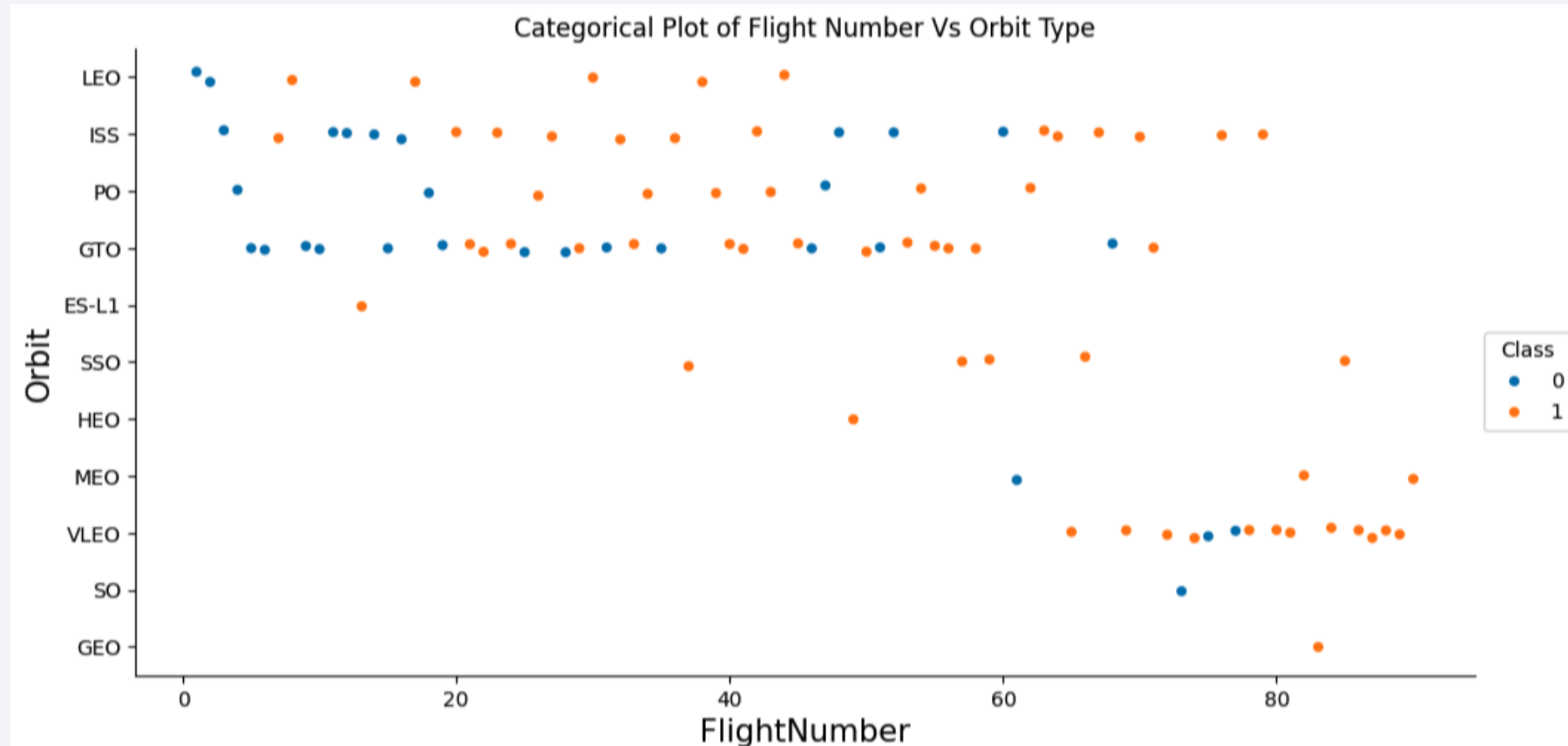
# Success Rate vs. Orbit Type

|                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>▪ VLEO(Very Low Earth Orbit) success rate - 85.7% (sample size 14)</li> </ul>                                  | <ul style="list-style-type: none"> <li>▪ LEO (Low Earth Orbit) success rate is 71.4% (sample size 7)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <ul style="list-style-type: none"> <li>▪ SSO(Sun-Synchronous Orbit) success rate - 100% (sample size of 5)<br/>* SSO is a type of LEO</li> </ul>      | <ul style="list-style-type: none"> <li>▪ PO(Polar Orbit) success rate - 66.7% on sample size 9<br/>* PO is a type of LEO</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <ul style="list-style-type: none"> <li>▪ ISS(International Space Station) success rate - 61.9% (sample size of 21)<br/>* ISS also a LEO</li> </ul>    | <ul style="list-style-type: none"> <li>▪ GTO (Geostationary Transfer Orbit) success rate - 51.8% (sample size of 27).<br/>* Note - this is not the final orbit but intermediate used to transfer the satellite from LEO to GEO.</li> </ul>                                                                                                                                                                                                                                                                                                                          |
| <ul style="list-style-type: none"> <li>▪ MEO(Medium Earth Orbit) success rates - 66.7% (sample size of 3)<br/>* region between LEO and GEO</li> </ul> | <ul style="list-style-type: none"> <li>▪ ES-L1(1st Lagrange point b/w Earth Sun) success rate 100%</li> <li>▪ GEO(Geostationary orbit) success rate 100%</li> <li>▪ HEO (Highly Elliptical Orbit) success rate 100%</li> <li>* Note: These orbits are farther away from Earth and launching rockets into these is in general more difficult and expensive.<br/>(Sample size is 1 for all 3 orbits) and all of them have success outcome.<br/>This doesn't mean its 100% likely to succeed in the future it just means we do not have sufficient samples.</li> </ul> |

❖ Missions targeting Lower Energy Orbits have higher chances of booster recovery.

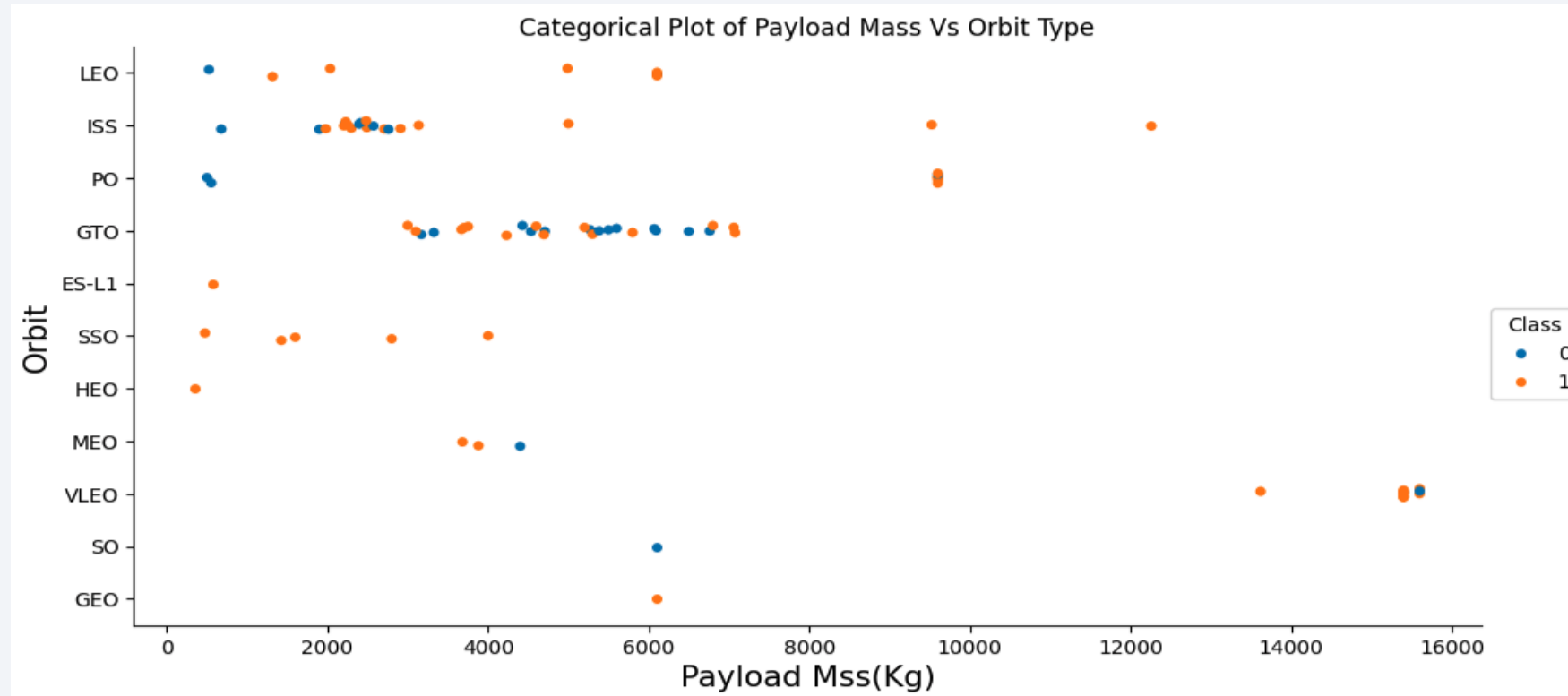


# Flight Number vs. Orbit Type



- ❖ LEO orbit - success seems related to flight #
- ❖ GTO - no relationship b/w flight # and success

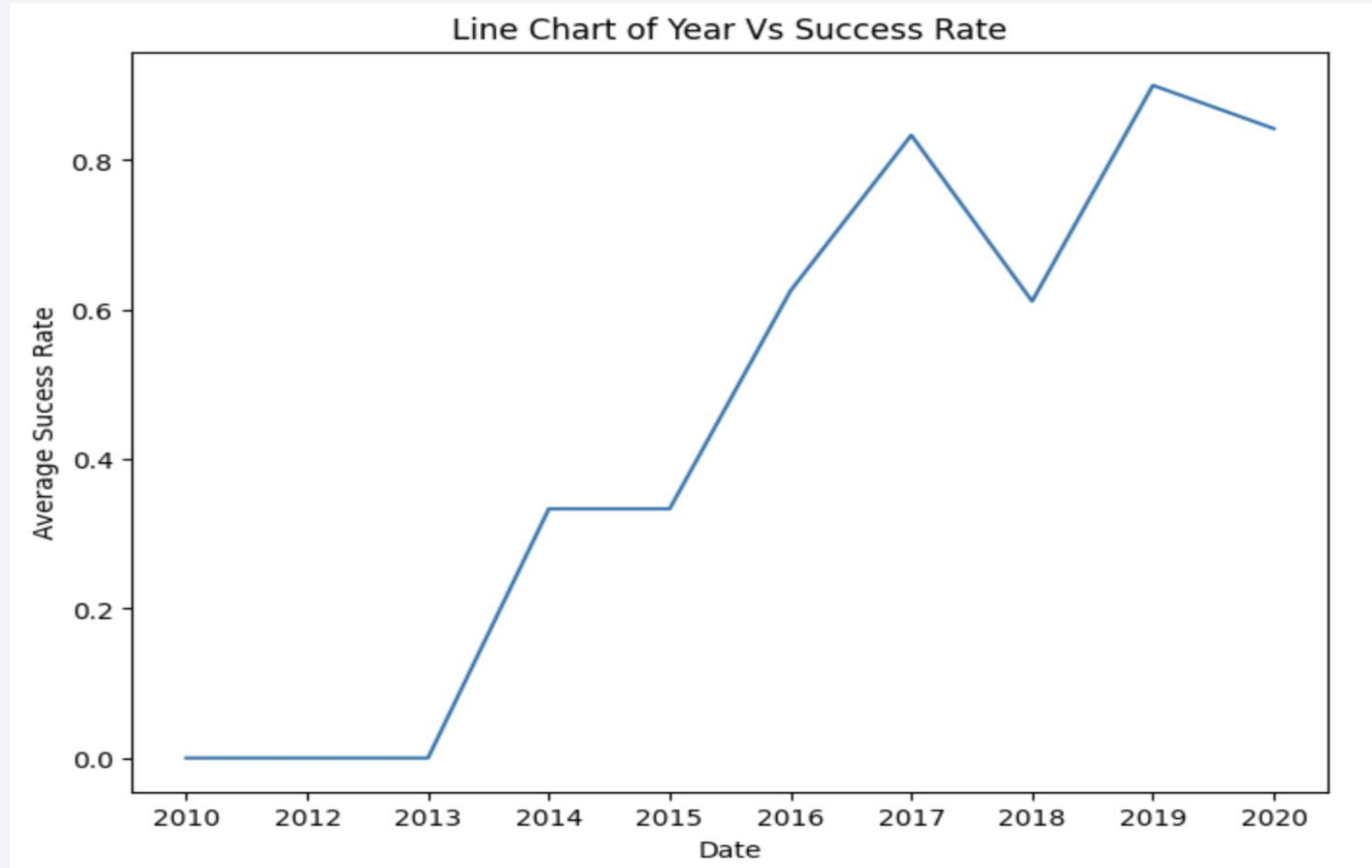
# Payload vs. Orbit Type



- **LEO, ISS AND PO** - high success rate for payloads  $> 4,000\text{kg}$  and more failures for payloads  $< 2,000\text{kg}$
- **ES-L1, SSO & HEO** - success with payload mass  $< 4000\text{kg}$
- **GTO** - hard to distinguish b/w successful and unsuccessful landings

# Launch Success Yearly Trend

Avg. Success  
Rate  
**INCREASING  
2013-2020.**



# All Launch Site Names

- Executed SQL query to select distinct values in the Launch\_Site column from SPACEXTBL table.
- Query Results

1. CCAFS LC-40
2. VAFB SLC-4E
3. KSC LC-39A
4. CCAFS SLC-40

```
%sql select distinct Launch_Site from SPACEXTBL
```

✓ 0.0s

\* [sqlite:///my\\_data1.db](#)

Done.

**Launch\_Site**

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

# Launch Site Names Begin with 'CCA'

- Executed SQL query that selects first 5 records from SPACEXTBL table where launch sites begin with 'CCA'

```
> %sql select * from SPACEXTBL where Launch_Site like 'CCA%' LIMIT 5
[16] ✓ 0.0s
... * sqlite:///my_data1.db
Done.
...

```

| Date       | Time (UTC) | Booster_Version | Launch_Site | Payload                                                       | PAYLOAD_MASS_KG_ | Orbit     | Customer        | Mission_Outcome | Landing_Outcome     |
|------------|------------|-----------------|-------------|---------------------------------------------------------------|------------------|-----------|-----------------|-----------------|---------------------|
| 2010-06-04 | 18:45:00   | F9 v1.0 B0003   | CCAFS LC-40 | Dragon Spacecraft Qualification Unit                          | 0                | LEO       | SpaceX          | Success         | Failure (parachute) |
| 2010-12-08 | 15:43:00   | F9 v1.0 B0004   | CCAFS LC-40 | Dragon demo flight C1, two CubeSats, barrel of Brouere cheese | 0                | LEO (ISS) | NASA (COTS) NRO | Success         | Failure (parachute) |
| 2012-05-22 | 7:44:00    | F9 v1.0 B0005   | CCAFS LC-40 | Dragon demo flight C2                                         | 525              | LEO (ISS) | NASA (COTS)     | Success         | No attempt          |
| 2012-10-08 | 0:35:00    | F9 v1.0 B0006   | CCAFS LC-40 | SpaceX CRS-1                                                  | 500              | LEO (ISS) | NASA (CRS)      | Success         | No attempt          |
| 2013-03-01 | 15:10:00   | F9 v1.0 B0007   | CCAFS LC-40 | SpaceX CRS-2                                                  | 677              | LEO (ISS) | NASA (CRS)      | Success         | No attempt          |

# Total Payload Mass

- Executed SQL Query to get Total payload(using SUM Aggregation) carried by boosters from NASA

```
▶ %sql select SUM(PAYLOAD_MASS_KG_) as total_payload from SPACEXTBL where Customer like '%NASA%CRS%'
[17] ✓ 0.0s
... * sqlite:///my_data1.db
Done.
...
total_payload
48213
```

❖ Total Payload Mass = 48,213Kg

# Average Payload Mass by F9 v1.1

- Executed SQL Query to get average payload mass(using AVG aggregation) from SPACEXTBL table where booster version is F9 v1.1.

```
> %sql select AVG(PAYLOAD_MASS_KG_) from SPACEXTBL WHERE Booster_Version like 'F9 v1.1'
18] ✓ 0.0s

.. * sqlite:///my\_data1.db
Done.

..
AVG(PAYLOAD_MASS_KG_)
2928.4
```

❖ Average Payload Mass carried by F9 v1.1 is 2928.4 Kg



# First Successful Ground Landing Date

- Executed SQL query to get values in Date column where the Landing Outcome is 'Success in Ground Pad', order the dates by ascending order and return the first date value.

```
> %sql select Date from SPACEXTBL where Landing_Outcome like '%Success%ground%pad%' order by Date ASC LIMIT 1
19] ✓ 0.0s

* sqlite:///my_data1.db
Done.

Date
2015-12-22
```

❖ First successful landing outcome on ground pad was on December 22, 2015.

## Boosters with Success in Drone Ship Landing with Payload range 4000-6000

- Executed SQL query to get Booster\_Version column values (without duplicates) where Landing\_Outcome is 'Success in Drone Ship' and payload mass greater than 4000 but less than 6000.

```
> %sql select distinct Booster_Version from SPACEXTBL where Landing_Outcome like '%Success%drone%ship%' and PAYLOAD_MASS_KG_ between 4000 and 6000
20] ✓ 0.0s
* sqlite:///my_data1.db
Done.
```

| Booster_Version |
|-----------------|
| F9 FT B1022     |
| F9 FT B1026     |
| F9 FT B1021.2   |
| F9 FT B1031.2   |

- Query Results:
  - F9 FT B1022
  - F9 FT B1021.2
  - F9 FT B1026
  - F9 FT B1031.2

# Total Number of Successful and Failure Mission Outcomes

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- Executed inner SQL to query SPACEXTBL table to return new column **outcome** where the value is 'SUCCESS' when Mission\_Outcome has the word 'success' and 'FAIL' otherwise.
- Then we aggregate the data (using SUM) returned by the inner SQL query

| outcome | COUNT(*) |
|---------|----------|
| FAIL    | 1        |
| SUCCESS | 100      |

- **Query Results:**

SUCCESS mission outcomes = 100  
FAIL mission outcome = 1

\* Note - Mission outcomes can be considered successful even for failed Landing\_Outcomes

# Boosters Carried Maximum Payload

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- Executed inner SQL query to get Maximum value of column Payload Mass from table SPACEXTBL
- Executed outer query to select Booster\_Version column values from table SPACEXTBL where the Payload Mass equals Max value calculated by inner query
- %sql SELECT Booster\_Version from SPACEXTBL WHERE PAYLOAD\_MASS\_\_KG\_ = (SELECT MAX(PAYLOAD\_MASS\_\_KG\_) from SPACEXTBL)
- Query Results:

|                 |                 |                 |                 |
|-----------------|-----------------|-----------------|-----------------|
| ▪ F9 B5 B1048.4 | ▪ F9 B5 B1048.5 | ▪ F9 B5 B1058.3 | ▪ F9 B5 B1049.4 |
| ▪ F9 B5 B1051.4 | ▪ F9 B5 B1051.6 | ▪ F9 B5 B1051.3 | ▪ F9 B5 B1049.5 |
| ▪ F9 B5 B1060.3 | ▪ F9 B5 B1056.4 | ▪ F9 B5 B1060.2 | ▪ F9 B5 B1049.7 |

# 2015 Launch Records

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- Executed SQL Query to get values of Date(Month & Year extracted), Landing\_Outcome, Booster\_Version and Launch\_Site columns from table SPACEXTBL filtered by year 2015 and where the Landing\_Outcome is 'failure (drone ship)'

```
%sql select substr(Date,6,2) as month_name, substr(Date,0,5) as yr,  
Landing_Outcome,Booster_Version, Launch_Site from SPACEXTBL where yr = '2015' AND  
Landing_Outcome like '%failure%drone%ship%'
```

- Query Results:

| month_name | yr   | Landing_Outcome      | Booster_Version | Launch_Site |
|------------|------|----------------------|-----------------|-------------|
| 01         | 2015 | Failure (drone ship) | F9 v1.1 B1012   | CCAFS LC-40 |
| 04         | 2015 | Failure (drone ship) | F9 v1.1 B1015   | CCAFS LC-40 |

# Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Executed SQL query on table SPACEXTBL to
  1. Get records dated between 2010-06-04 and 2017-03-20
  2. Group records by Landing\_Outcome
  3. Project Landing\_Outcome and its count using aggregation
  4. Sort the records in descending order of count.

```
%sql select Landing_Outcome, Count(*) from SPACEXTBL where Date  
between '2010-06-04' and '2017-03-20' GROUP BY Landing_Outcome  
ORDER by Date DESC
```

## Query Results:

| Landing_Outcome        | Count(*) |
|------------------------|----------|
| No attempt             | 10       |
| Success (drone ship)   | 5        |
| Failure (drone ship)   | 5        |
| Success (ground pad)   | 3        |
| Controlled (ocean)     | 3        |
| Uncontrolled (ocean)   | 2        |
| Failure (parachute)    | 2        |
| Precluded (drone ship) | 1        |

# EDA - Results Summary

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- As flight # increases (continuous launch attempts) - first stage is more likely to land successfully.
- 'VAFB SLC 4E' and 'KSC LC 39A' sites have higher probability of successful landing
- 'VAFB SLC 4E' - No rockets were launched with payloads > 10,000 kg.
- 'CCAFS SLC 40' - seems good at launching rockets w/ payloads > 10,000kg - however sample size is too small.
- 'KSC LC 39A' - very good success rate with wide range of payload mass b/w 2,000 –16000 Kg.
- Missions targeting Low Earth Orbits like VLEO, LEO, SSO, PO have higher chance of booster recovery.
- Successful landing rate is more for low energy orbits like PO, LEO & ISS & VLEO even with heavier payloads
- First Successful landing outcome in ground pad was achieved in Dec 22, 2015
- Boosters that carried maximum payload mass were Flacon 9 Block 4 type

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a curved line separating the dark surface from the deep blue of space.

Section 3

# Launch Sites Proximities Analysis



# Launch Sites Markers on Folium Map

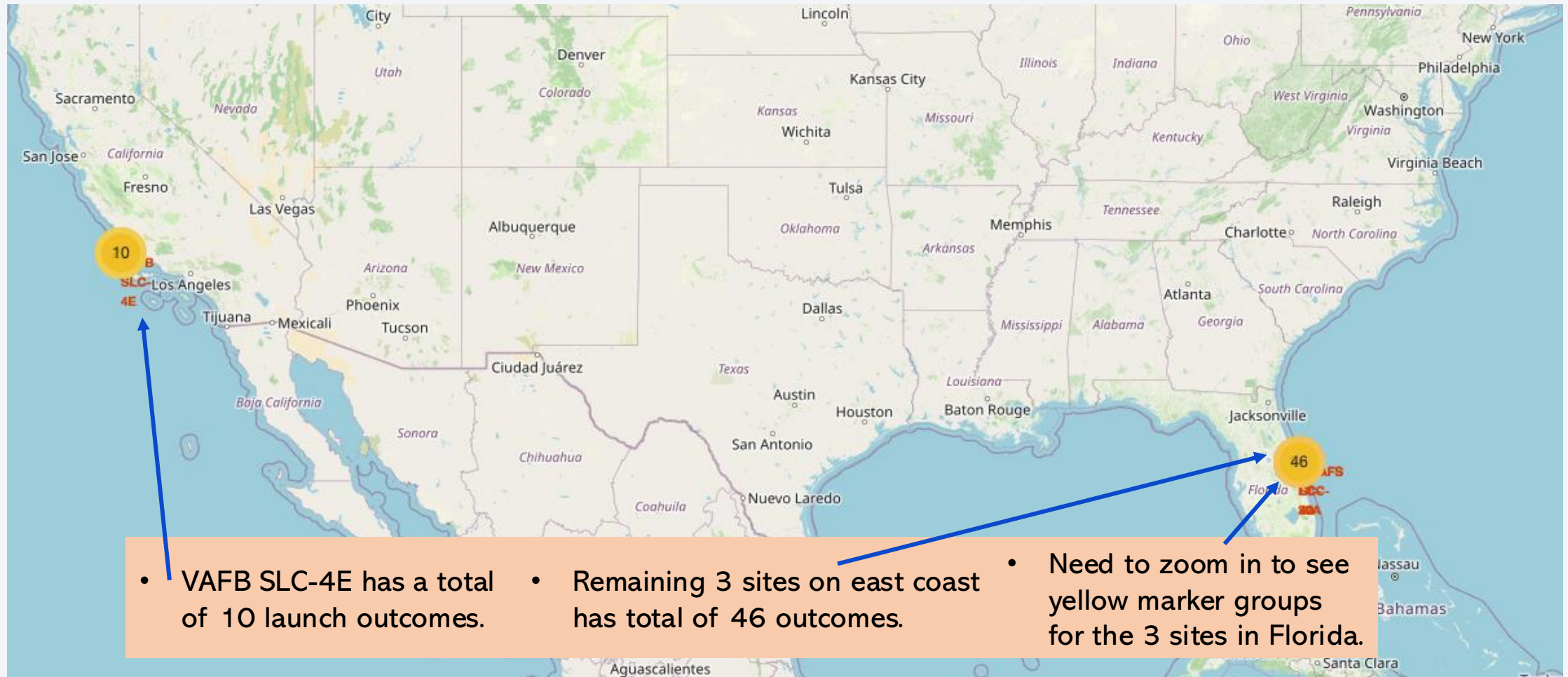


Zoomed in  
view



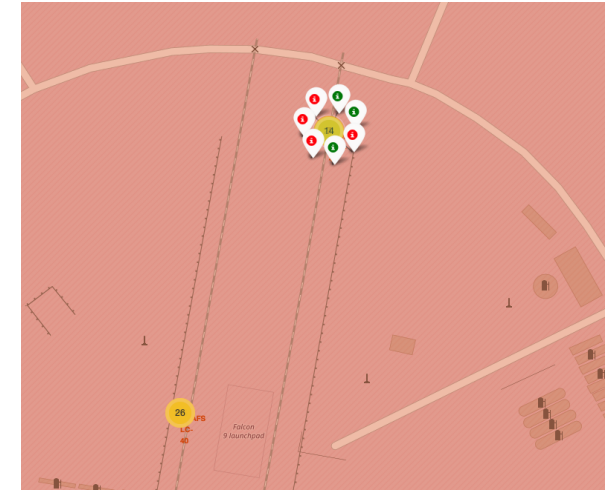
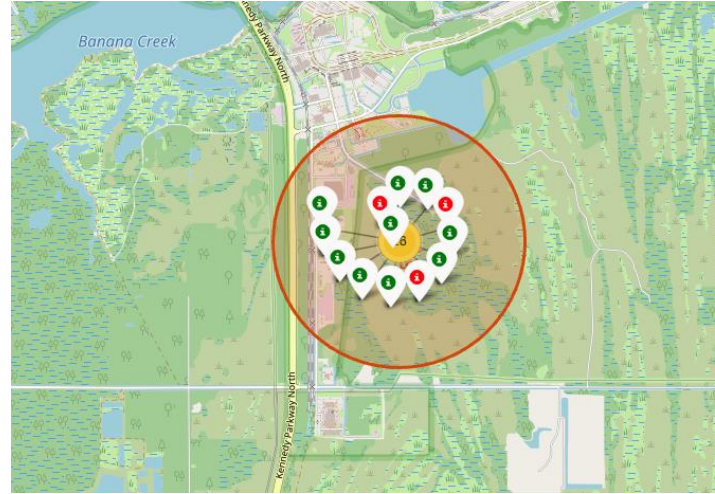
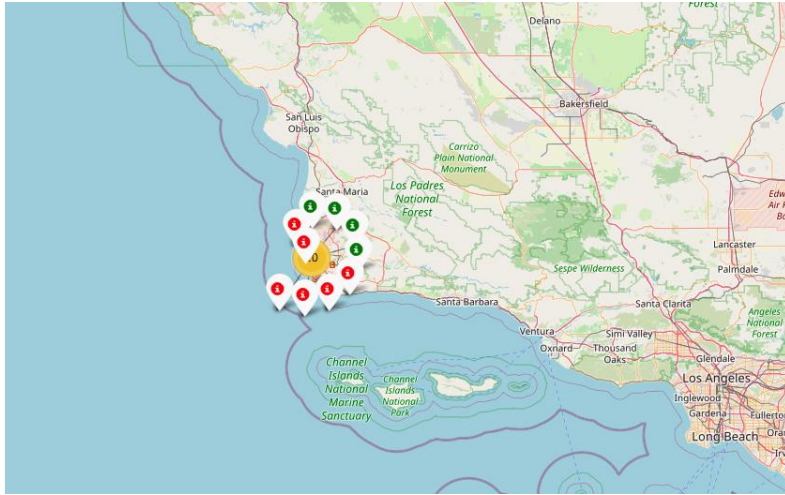
All launch sites are  
in close proximity to  
the equator and on  
the coast

# Folium Map with Marker Clusters for Launch Sites

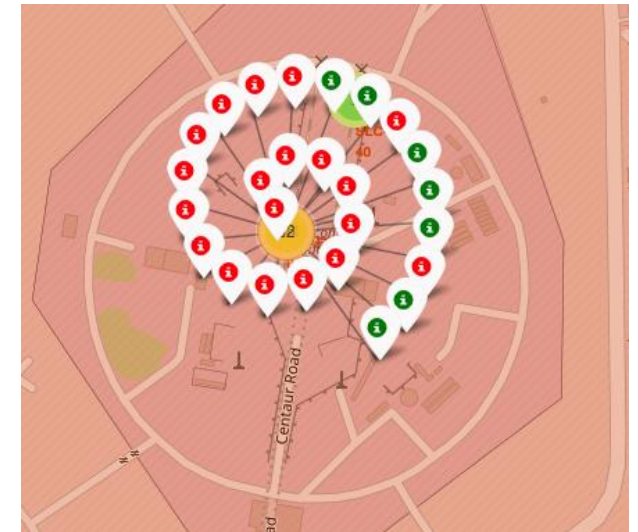




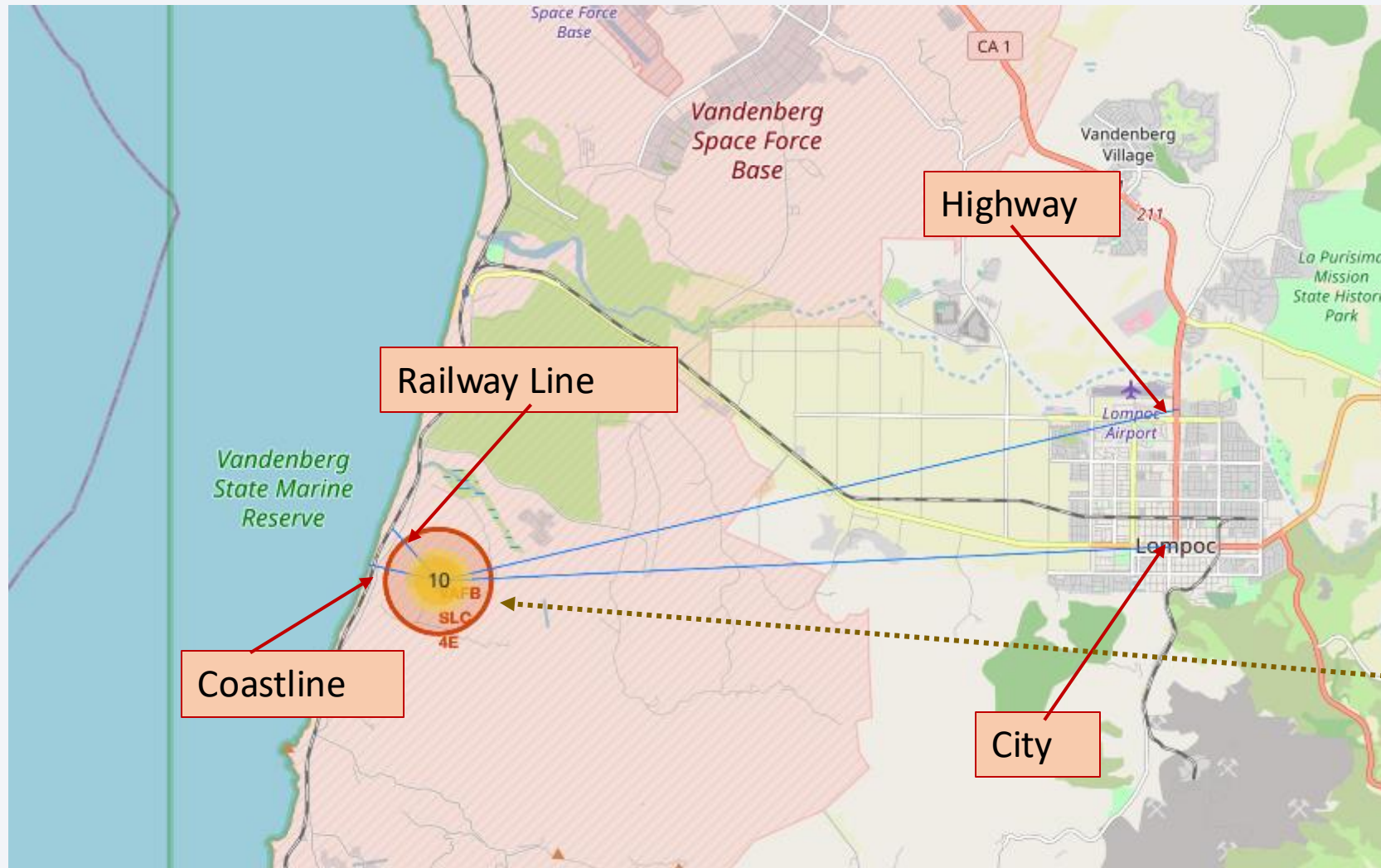
# Success vs Failures for each Launch Site



- Clicking on yellow group marker shows how many passed and failed
- VAFB SLC-4E Success Rate of 4/10
- KSC LC-39A Success Rate 10/13
- CCAFS SLC-40 & CCAFS LC-40 have success rates of 3/7 & 7/26 respectively
- **KSC LC-39A has the highest chances of SUCCESS** followed by CCAFS SLC-40 & VAFB SLC-4E
- **CCAFS LC-40 has the lowest chance**



# Folium Map - Launch Site 'VAFB SLC-4E'



Launch site 'VAFB SLC-4E' is in close proximity to Railway line, coast line, highway and city

# Launch Sites Proximities Analysis - Results Summary

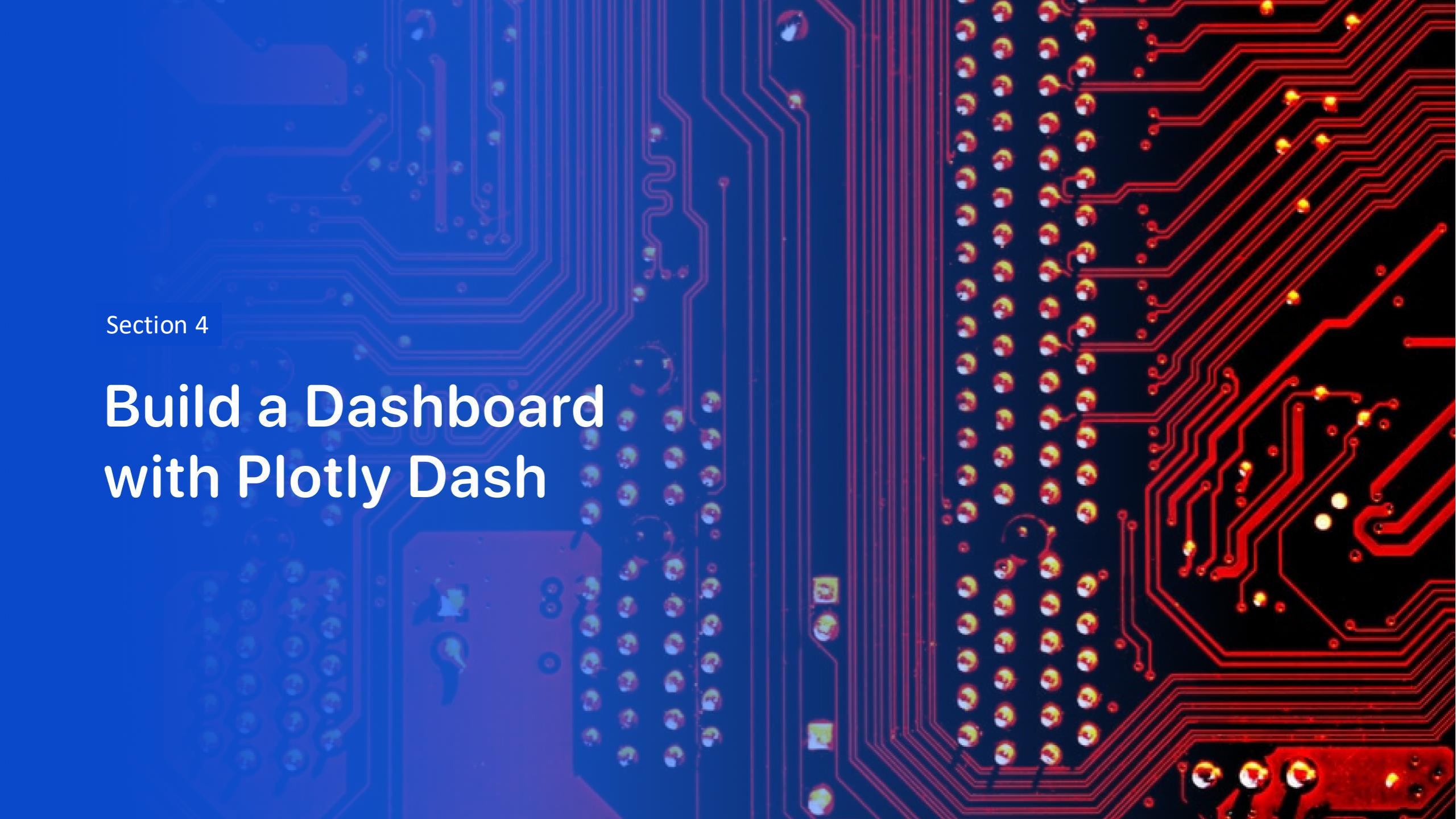
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- 4 Major Launch Sites in US - 1 on West coast CA and 3 on the east coast FL
- Launch sites are in close proximity to Equator and Ocean.

| Launch Site  | Success | Fail | Success rate |
|--------------|---------|------|--------------|
| KSC LC-39A   | 10      | 3    | 76.9%        |
| CCAFS SLC-40 | 3       | 4    | 42.8%        |
| VAFB SLC-4E  | 4       | 6    | 40%          |
| CCAFS LC-40  | 7       | 19   | 26.9%        |

- KSC LC-39A has the highest chances of launch success rate
- Launch sites are in close proximity to key infrastructure facilities like - coast line, railway line, city & highway.

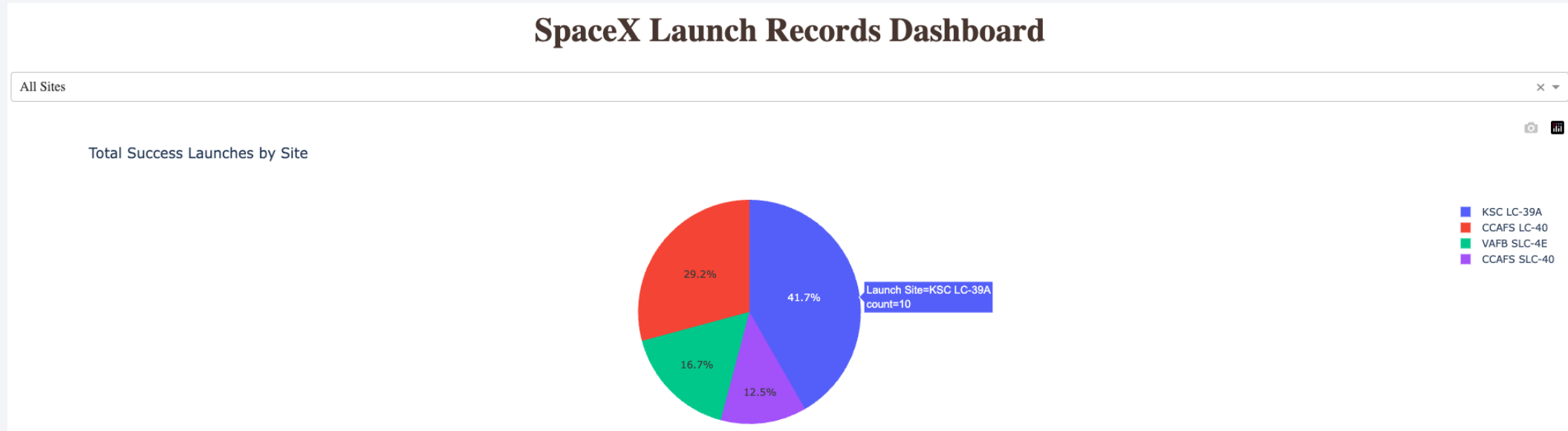




Section 4

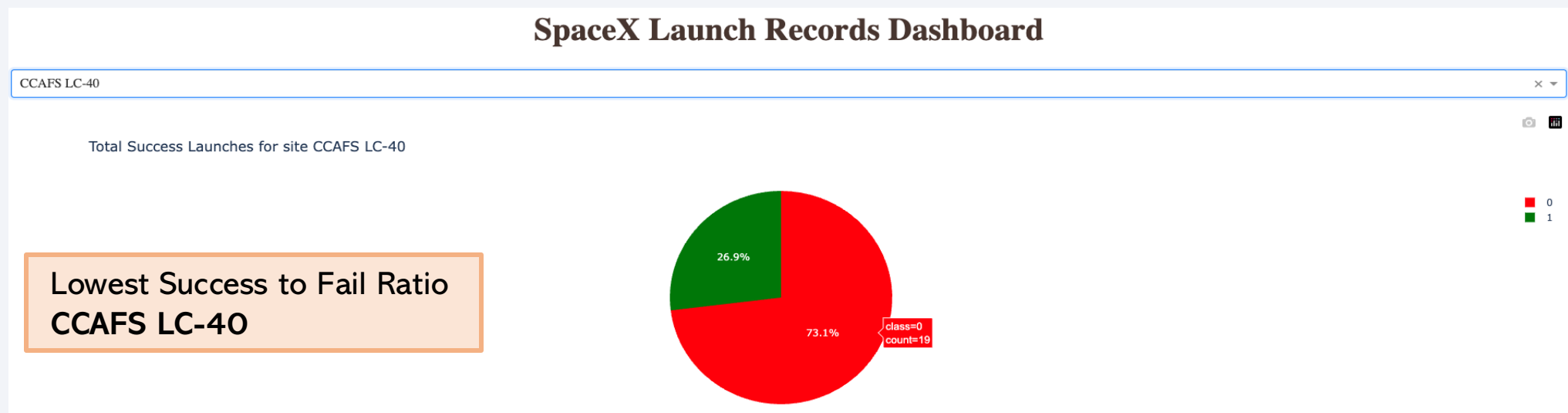
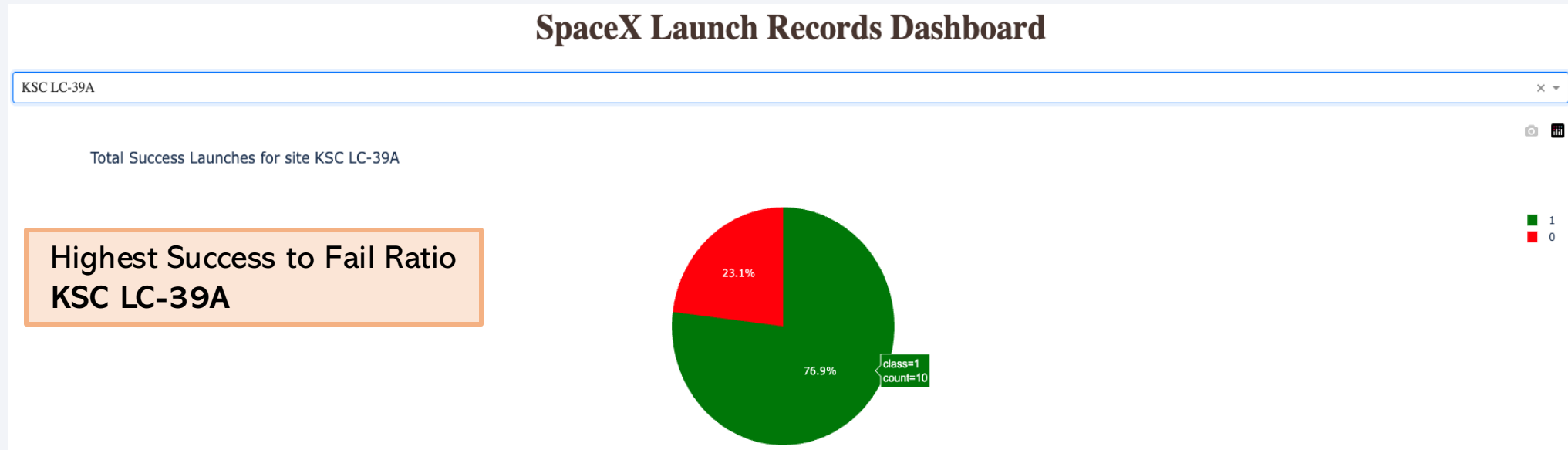
# Build a Dashboard with Plotly Dash

# Dashboard – Pie Chart Of Total Success Launches by Site



- ❖ KSC LC-39A has the highest share(41.7%)
- ❖ CCAFS SLC-40 has lowest (12.5%)

# Dashboard – Pie Chart Of Successes vs Failures for site 'KSC LC-39A'





# Dashboard – Plot of Payload Mass vs Class

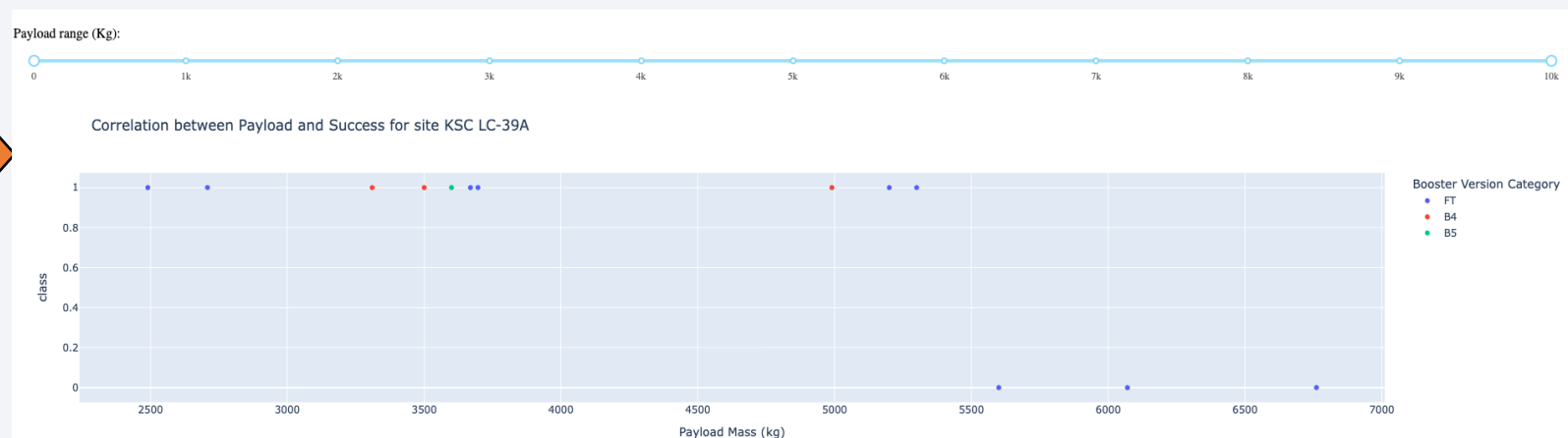
Payload Range

0 – 10,000 (Kgs)

- Booster Version v1.0-1.1 performs worst.
- FT performs best followed by B4 & B5

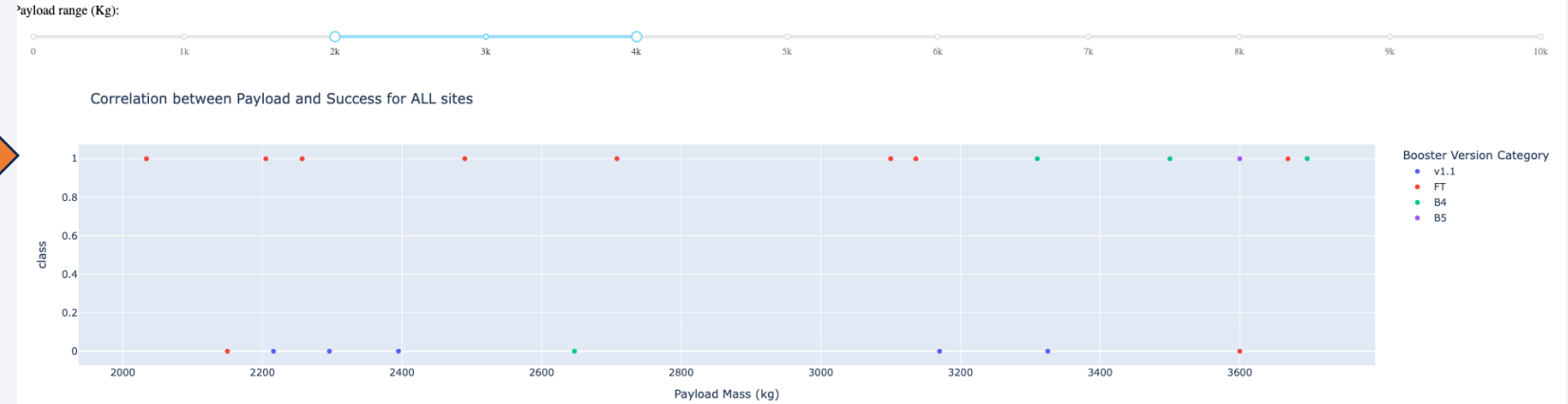


- Heavier payloads in general decrease chances of booster recovery.
- We can see this trend clearly for KSC LC-39A payloads > 5500kgs fail

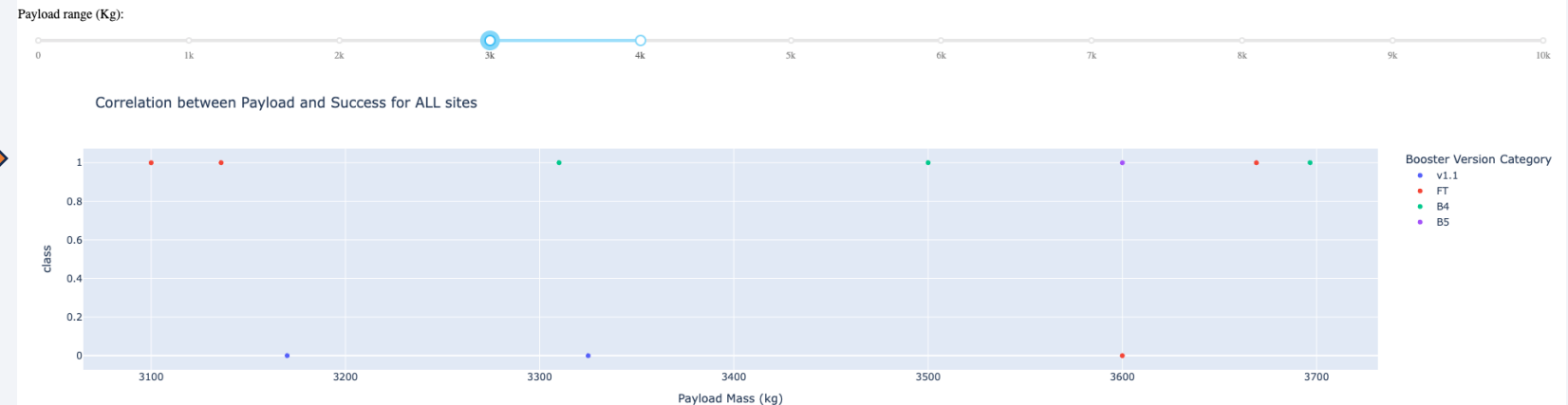


# Dashboard – Plot of Payload Mass vs Class

**Payload Range  
2K-4K has 60%  
success rate.**



**Payload Range  
3K-4K has 70%  
success rate.**



# Dashboard Visual Analytics - Results Summary

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- Site with the largest successful launches
  - **KSC LC-39A**
- Site with the highest launch success rate?
  - **KSC LC-39A with success rate of 76%**
- Payload range(s) having highest launch success rate
  - **3K-4K range has success rate of 70%**
  - 2K-4K range has success rate of 60%
  - 2K-5K range has success rate of 53%
- Payload range(s) having lowest launch success rate
  - **6K-10K has the lowest success rate of 16%**
- F9 Booster version (v1.0, v1.1, FT, B4, B5, etc.) having highest launch success rate?
  - **FT seems to have the highest launch success rate followed by B4 and B5**
  - Note: We do not have sufficient samples for B5 the only sample is a success



Section 5

# Predictive Analysis (Classification)

# Classification Accuracy

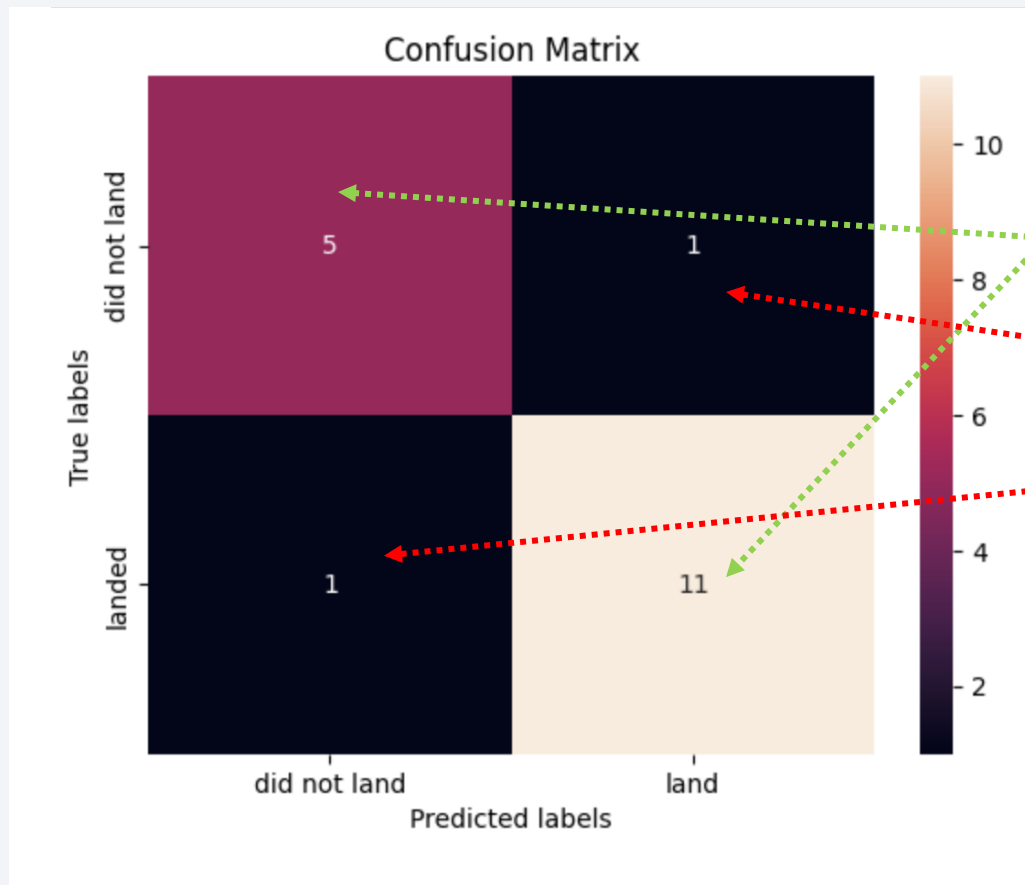
Results of Predictive Analysis

| Type                      | Tuned Hyperparameters                                                                                                                  | Accuracy Score<br>(Training set) | Accuracy Score<br>(Testing set) |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|---------------------------------|
| Logistic Regression       | 'C': 0.01, 'penalty': 'l2', 'solver': 'lbfgs'                                                                                          | 0.846                            | 0.833                           |
| Support Vector<br>Machine | 'C': 1.0, 'gamma':<br>0.03162277660168379, 'kernel':<br>'sigmoid'                                                                      | 0.848                            | 0.833                           |
| Decision Tree Classifier  | criterion': 'gini', 'max_depth': 6,<br>'max_features': 'sqrt',<br>'min_samples_leaf': 4,<br>'min_samples_split': 2, 'splitter': 'best' | 0.875                            | 0.888                           |
| K Neighbors Classifier    | 'algorithm': 'auto', 'metric': 'manhattan',<br>'n_neighbors': 10, 'p': 1, 'weights':<br>'uniform'                                      | 0.848                            | 0.833                           |

❖ Decision Tree Classifier model performed best on testing set with an accuracy score of 0.888

# Confusion Matrix

## Decision Tree Classifier



- True Positive – 11 (True label is landed, Predicted label is also landed)
- False Negative – 5 (True label is not landed, predicted label is not landed)
- False Positive – 1 (True label is not landed, Predicted label is landed)
- True Negative – 1 (True label is landed and predicted label is not landed)

❖ Only 2 out of 18 predictions went wrong

# Predictive Analysis - Results Summary

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- Decision Tree Classifier Model performed better than its counterparts on both training & testing sets -
  - Testing set - accuracy score of 0.888
  - Training set - accuracy score of 0.875
- All 3 models (Logistic Regression, SVM & KNN) performed with nearly same accuracy scores on both testing and training sets but lower than DTC
  - Testing set - accuracy score of 0.833
  - Training set - accuracy score of  $\sim 0.846$



# Conclusions

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- Our feasibility study for SpaceY's booster recovery demonstrates a robust predictive framework.
- By leveraging cleaned and meticulously prepared SpaceX launch data, we identified key features affecting booster recovery and developed multiple classification models.
- Among these, the Decision Tree Classifier consistently achieved an impressive accuracy of approximately 89%, indicating strong reliability in predicting booster recovery outcomes.
- This model not only supports launch success predictions but also holds potential for integration into cost estimation processes, enhancing overall operational efficiency for SpaceY.

# Appendix

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- GitHub Repository Link - [DataScience-Capstone](#)
- Data Collection through API Notebook - [jupyter-labs-spacex-data-collection-api.ipynb](#)
- Data Collection through Web scraping Notebook - [jupyter-labs-spacex-data-collection-api.ipynb](#)
- Data Wrangling Notebook - [labs-jupyter-spacex-Data wrangling.ipynb](#)
- EDA with Visualization Notebook - [edadataviz.ipynb](#)
- EDA with SQL Notebook - [jupyter-labs-eda-sql-coursera\\_sqlite.ipynb](#)
- Interactive Visual Analytics with Folium Lab Notebook - [lab\\_jupyter\\_launch\\_site\\_location.ipynb](#)
- Visual Analytics Dashboard Code - [spacex-dash-app.py](#)
- Machine Learning Prediction Notebook - [SpaceX Machine Learning Prediction Part 5.ipynb](#)

Thank you!

